QUANTUM MECHANICS

General Chemistry I, Lecture Series 5

Qixi Mi

Reading:
OGC8 §4

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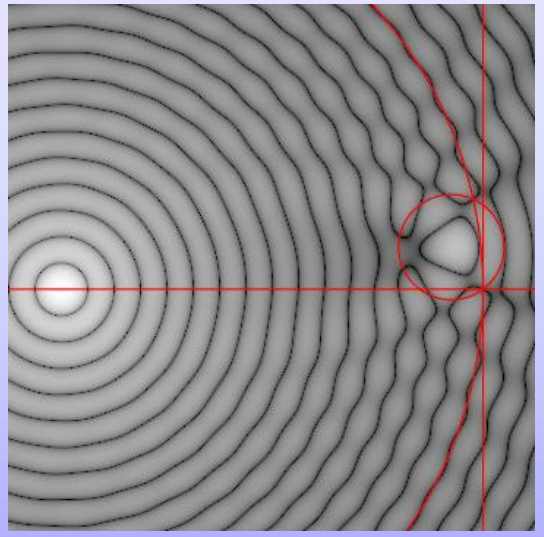
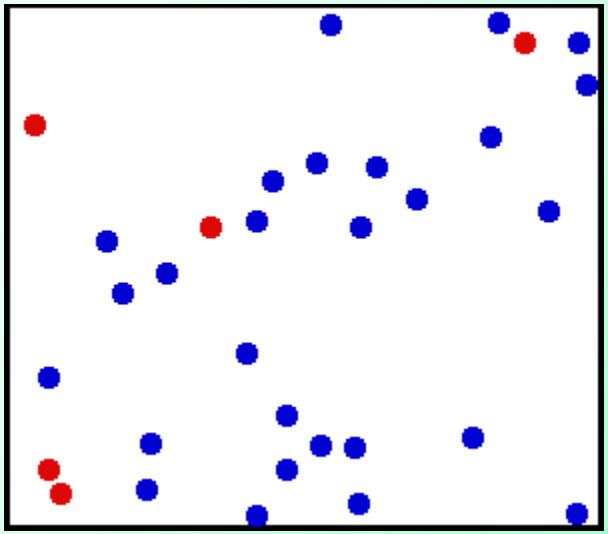
Outline

- The particle nature of light
  - Photon
  - The uncertainty principle
- The wave nature of electron
  - Wave motion
  - Wave function

Wave vs. Particle

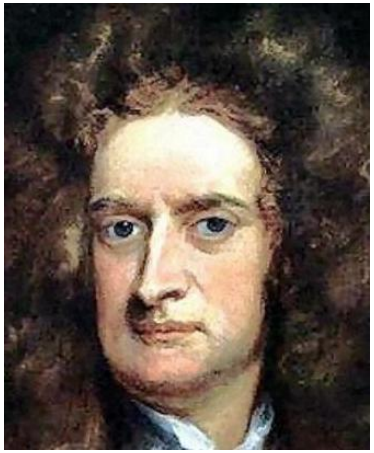


- ✓ Wavelength
- ✓ Frequency
- ✓ Interference
- ✓ Diffraction



- ✓ Energy
- ✓ Momentum
- ✓ Trajectory 轨迹

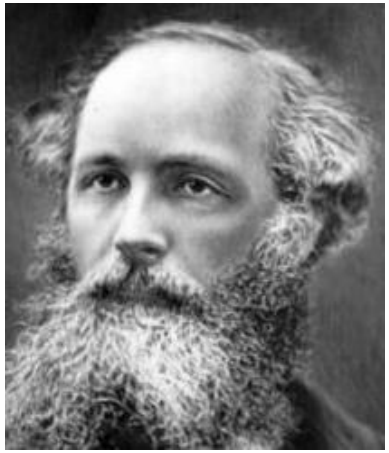
Light: Wave or Particle? (1)



Isaac Newton
(1643–1727)

VS

Christiaan Huygens
(1629–1695)



James C. Maxwell
(1831–1879)

Light: Wave or Particle? (2)

Max Planck
(Berlin, 1858–1947)

Niels Bohr
(Copenhagen, Cambridge
1885–1962)

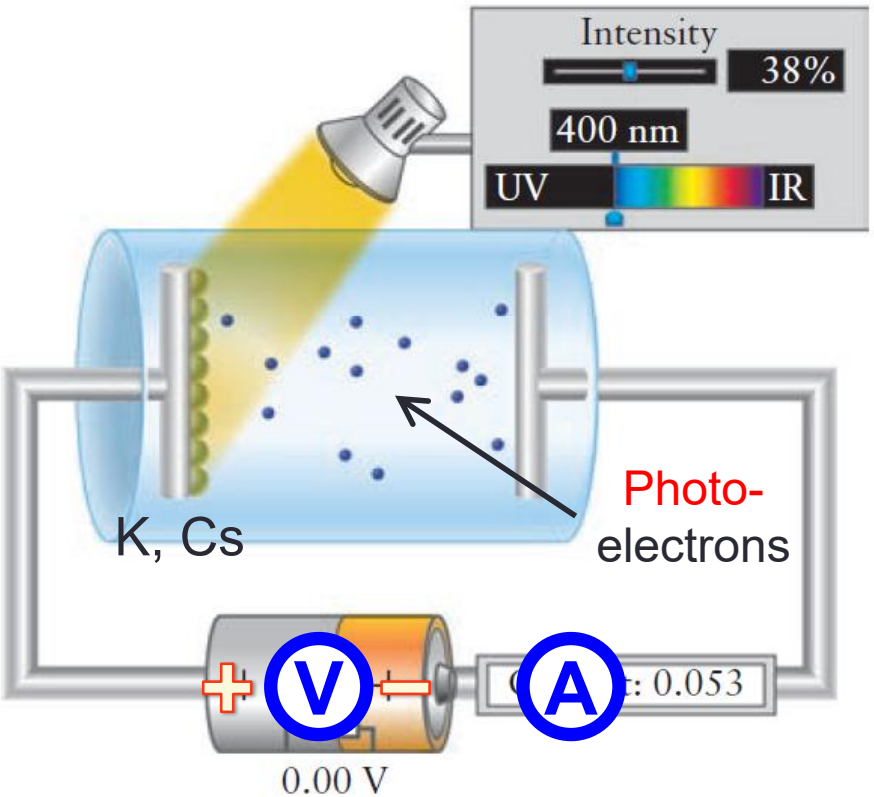
Albert Einstein
(Zürich, 1879–1955)

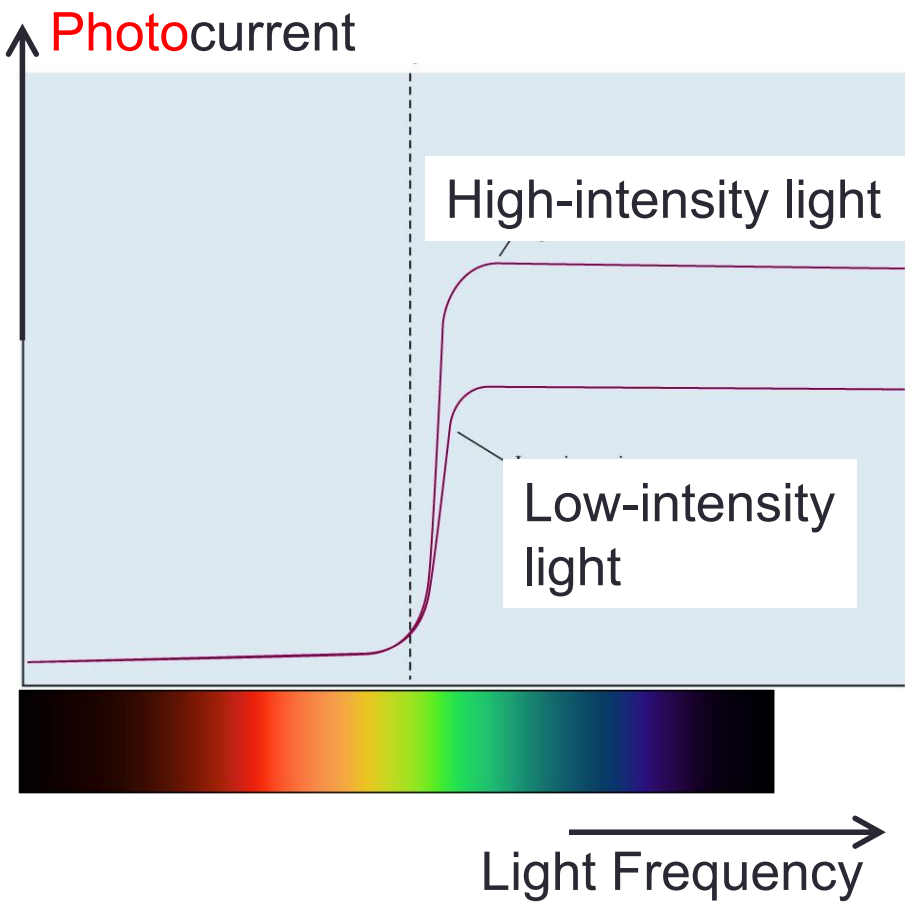
"Energy is quantized in matter but not in light."

"Light is also quantized."

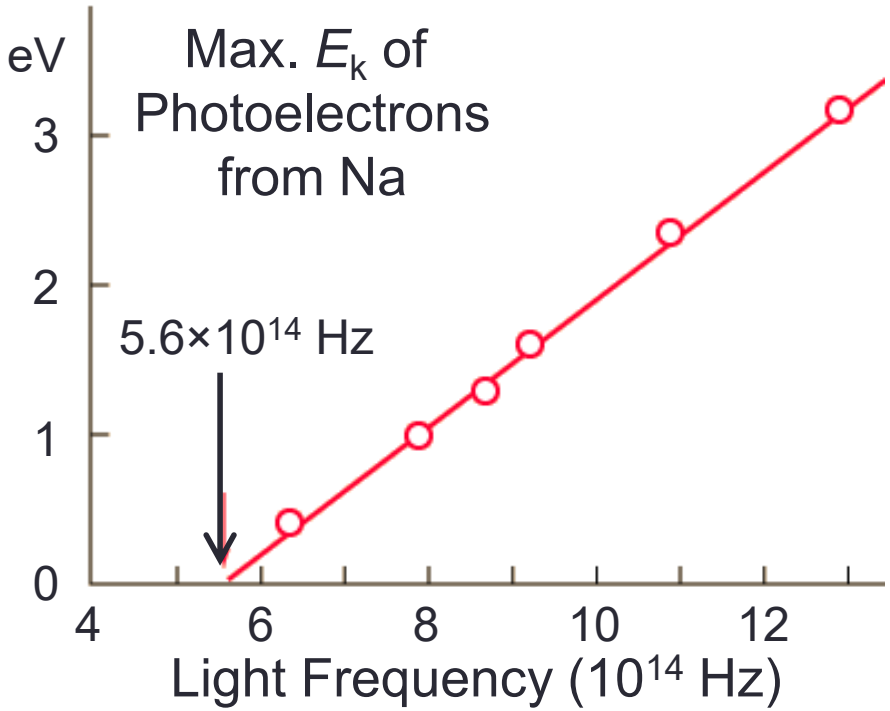
The Photoelectric Effect

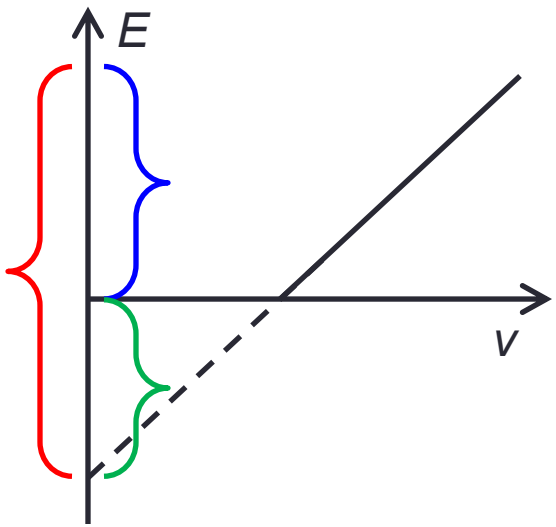
Photo- 光

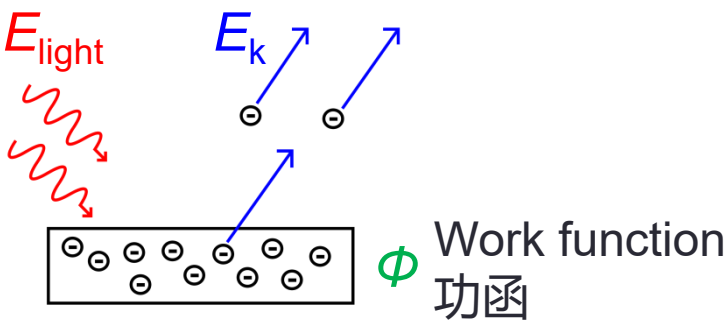




Photon: The Energy Quanta of Light







Einstein 1905:

$$E_{k,max} + \Phi = E_{light} \propto v$$

$$E_{light} = hv$$

Light Also Bears Momentum

(1) Planck 1900, Einstein 1905:

$$E = h\nu$$

(2) Einstein 1916:

$$E = mc^2$$

(3) Definition of momentum

$$p = mv, \quad v = c \text{ for light}$$

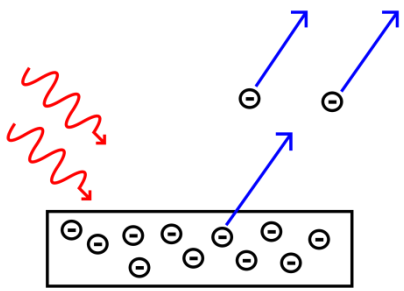
(1)+(2)+(3)

$$\boxed{p = \frac{h\nu}{c} = \frac{h}{\lambda}}$$

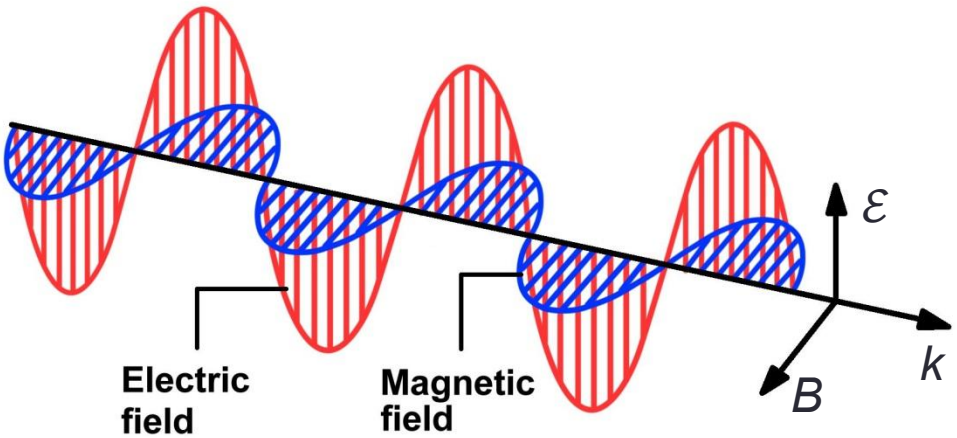
Photo- + -on = Photon 光子

Albert Einstein
(Zürich, 1879–1955)

Photon and Electromagnetic Wave
电磁波



VS



Photoelectric effect
(1) $E_{photon} = h\nu$

Electromagnetic wave
(2) $E_{light} \propto \mathcal{E}^2$

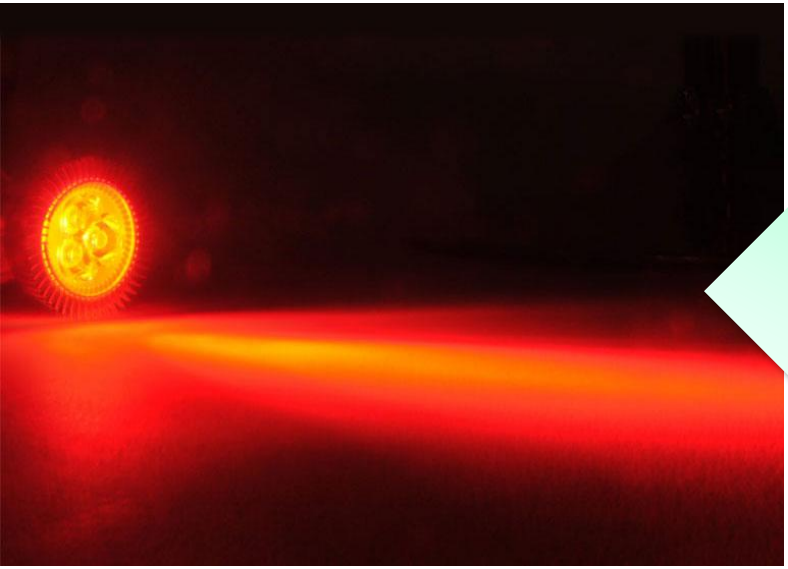
(1)+(2):

$N_{photon} \propto \mathcal{E}^2$

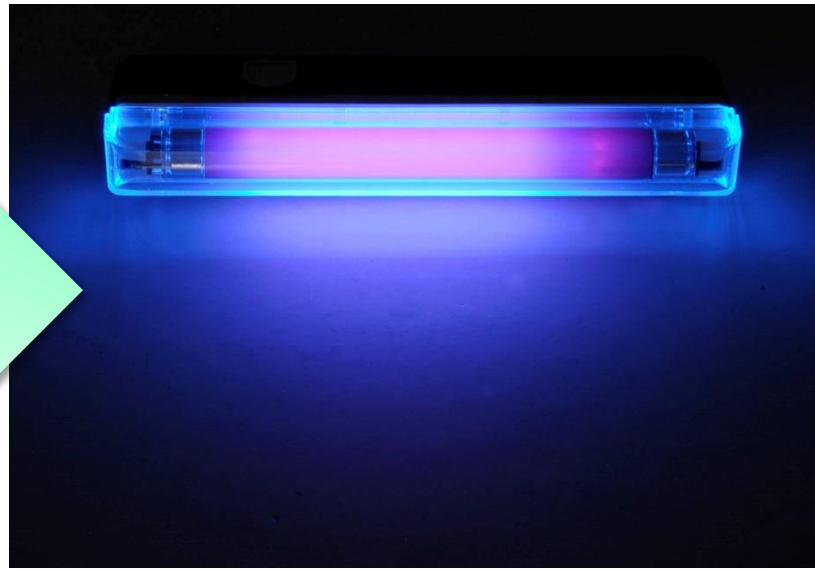
Light frequency → Energy of a single photon

Light intensity → Number of photons

Light and Rain: An Analogy



Equal Intensity



Equal precipitation

The Uncertainty Principle

For photon, $p = \frac{h\nu}{c} = \frac{h}{\lambda}$

For electron, $p = m_e v = \frac{h}{\lambda}$

Common trend:

$$p\lambda = h$$

Werner Heisenberg (Copenhagen, Leipzig, 1901–1976)

Heisenberg 1927:

$$\Delta p \cdot \Delta x \geq \frac{h}{4\pi}$$

Standard deviation
标准差

Survey

Which of the following is INCORRECT about the Heisenberg Uncertainty Principle?

A. The position *x* and momentum *p* of a particle cannot be known precisely at the same time.

B. The angular momentum *L* of a particle cannot be known precisely.

C. The limit of uncertainty has the dimension of J·s.

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- The particle nature of light
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  - The uncertainty principle
- The wave nature of electron
  - Wave motion
  - Wave function

The Wavelength of Electron

(1) Bohr 1913:

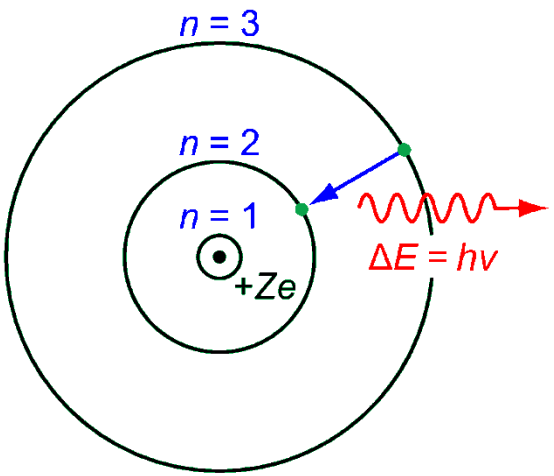
$$L = m_e v r = n\frac{h}{2\pi}$$

(2) de Broglie 1924:

$$2\pi r = n\lambda$$

(1)+(2)

$$p = m_e v = \frac{h}{\lambda}$$

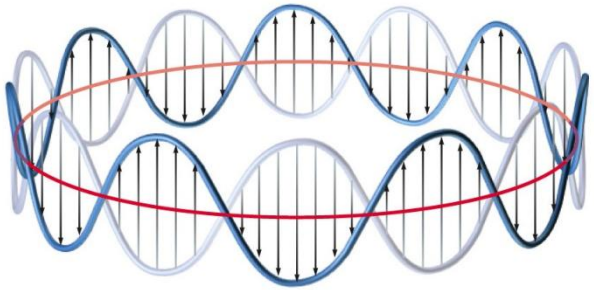


Louis de Broglie (Sorbonne, 1892–1987)

Stationary state 定态

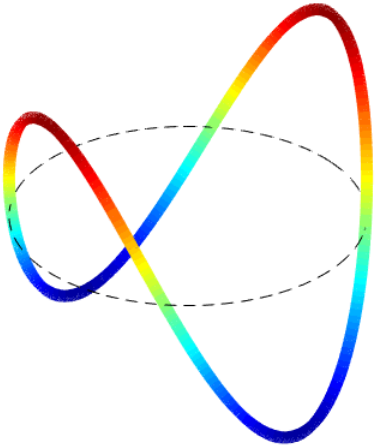
↕

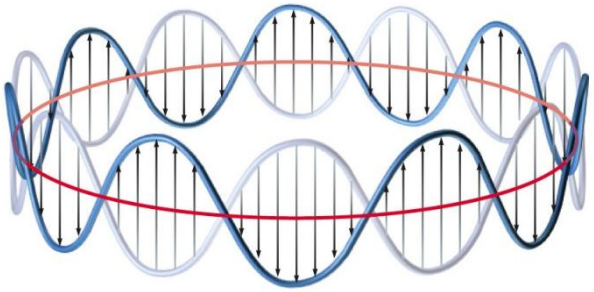
Standing wave 驻波

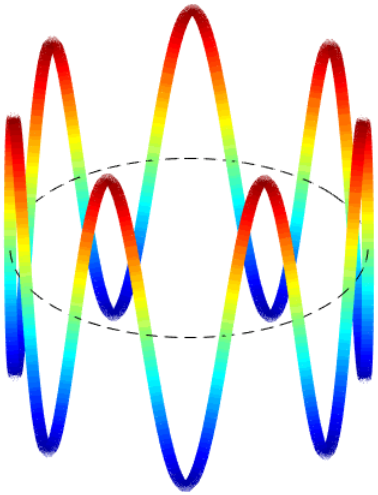


$2\pi r = 7\lambda$

DEMO: Circular Standing Waves

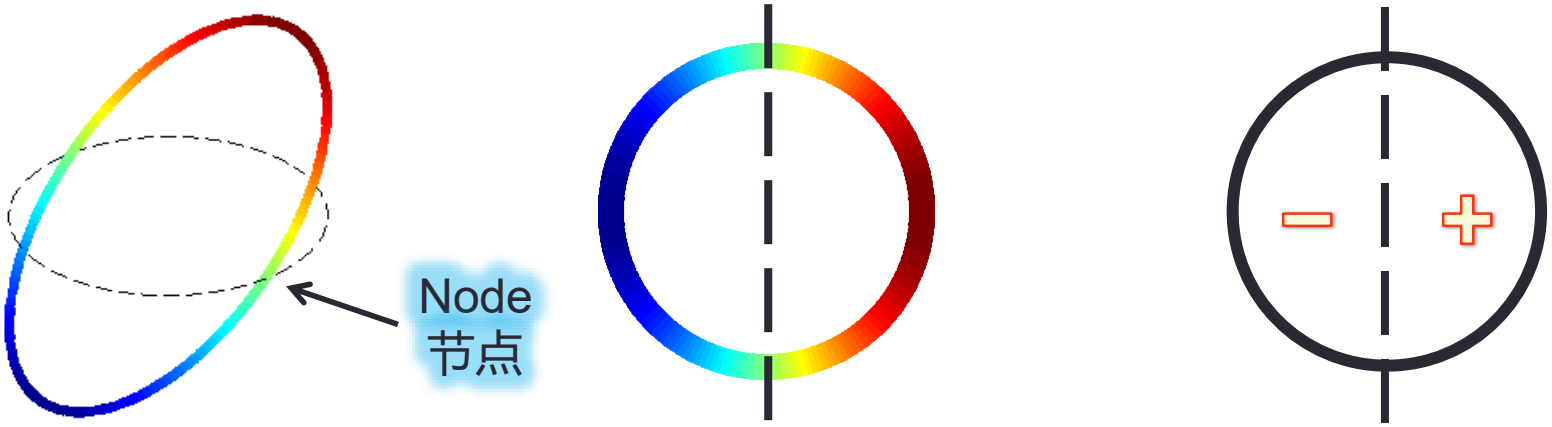




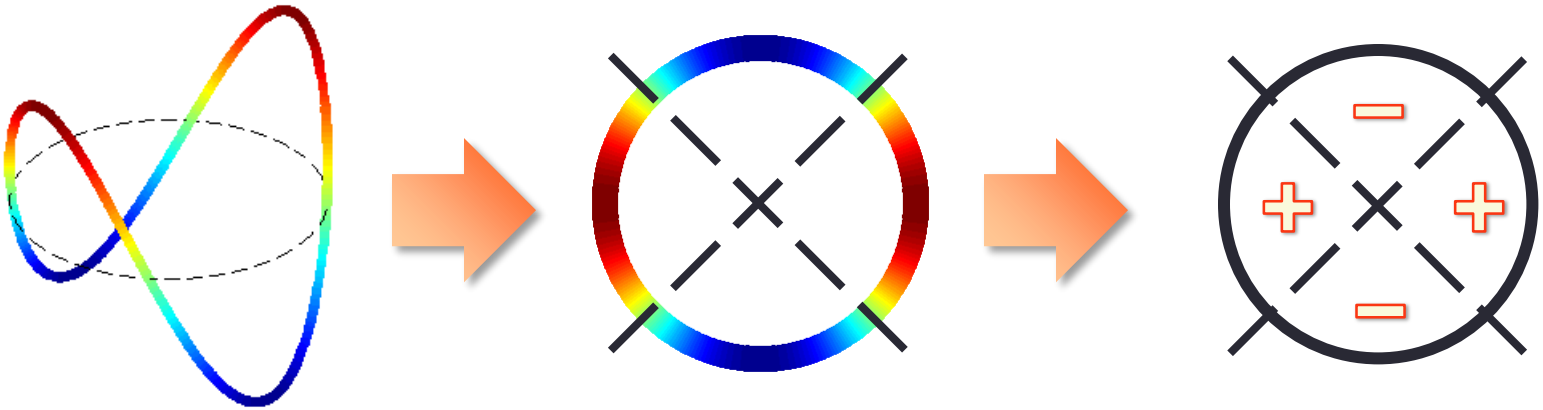


de Broglie wave

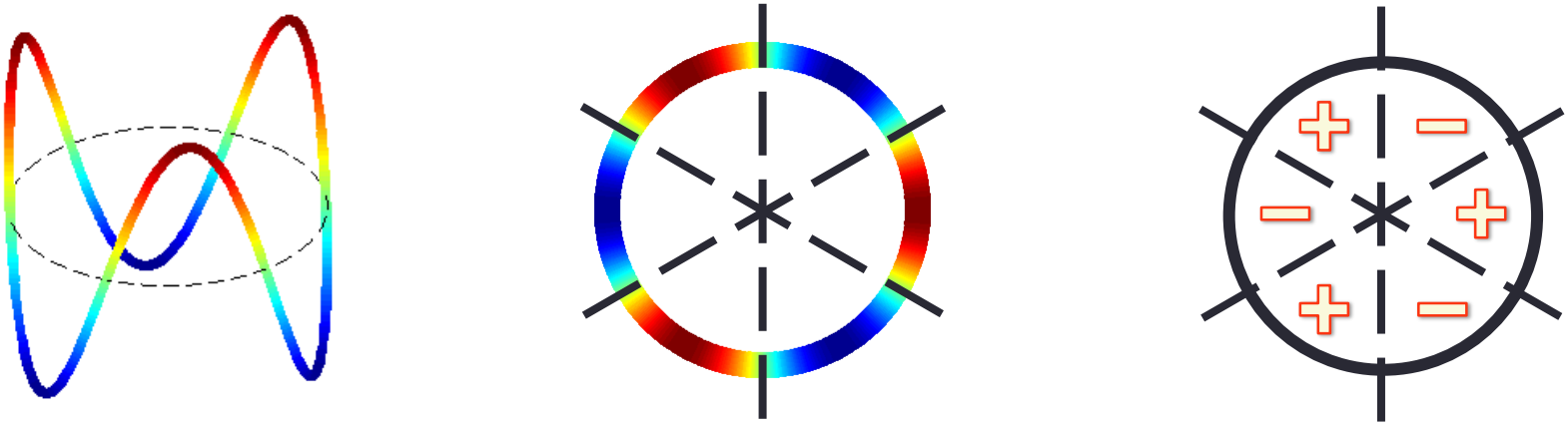
$2\pi r = \lambda$



$2\pi r = 2\lambda$



$2\pi r = 3\lambda$



Outline

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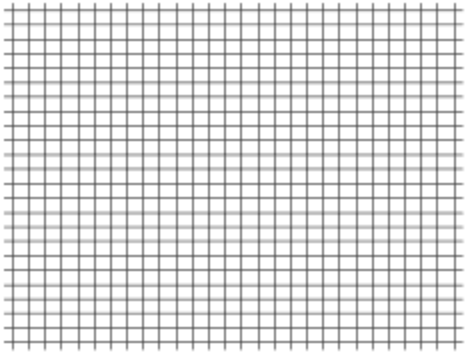
- The wave nature of electron
  - Wave motion
  - Wave function

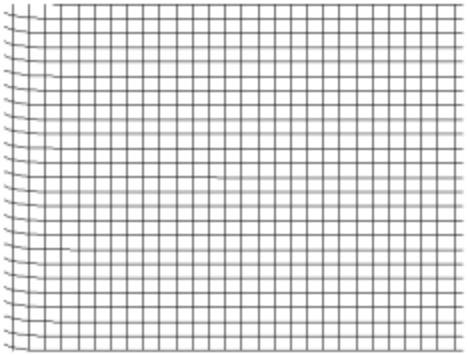
Wave: Basics

Longitudinal wave
纵波

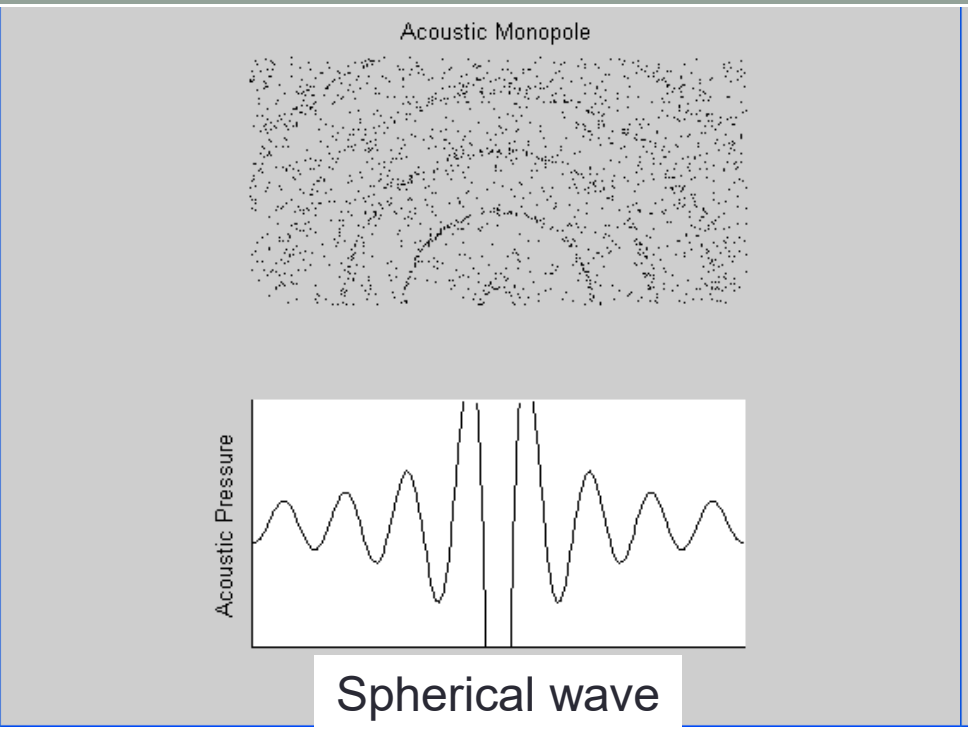


Direction of propagation

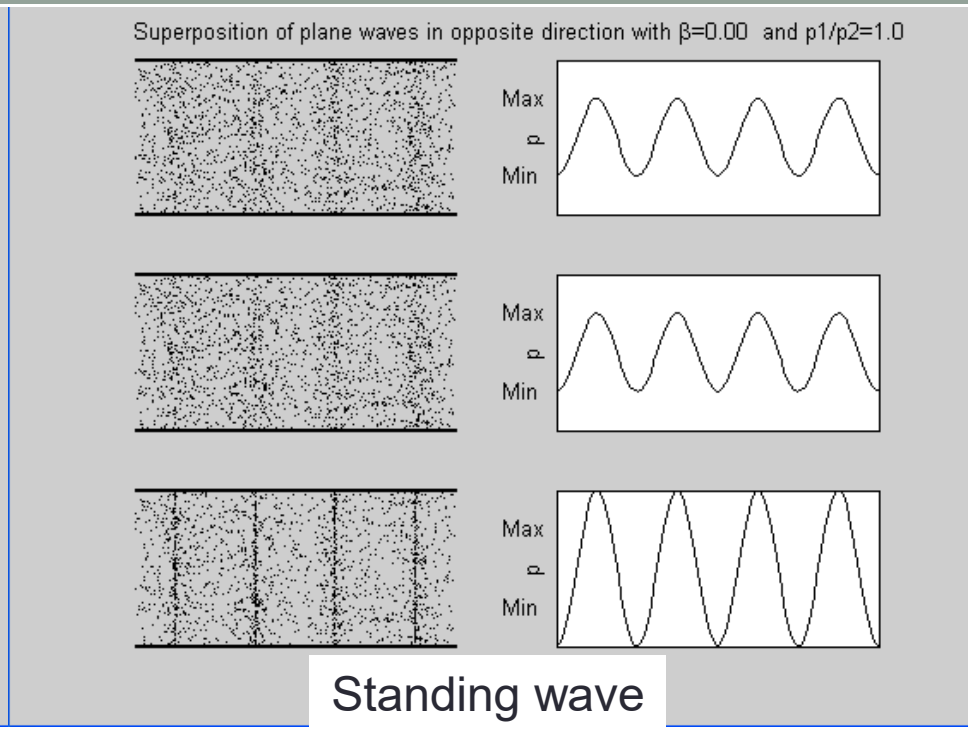
Transverse wave
横波



- Wave
  - Propagation 传播
  - Oscillation 振荡
    - Longitudinal
    - Transverse

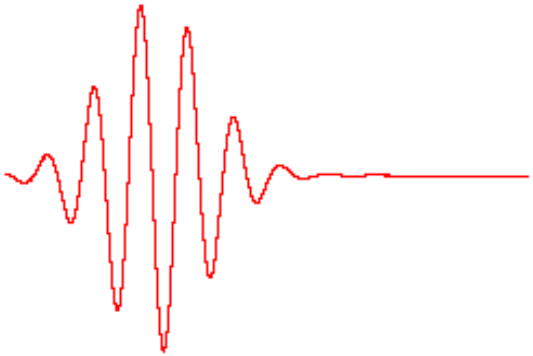


Spherical wave

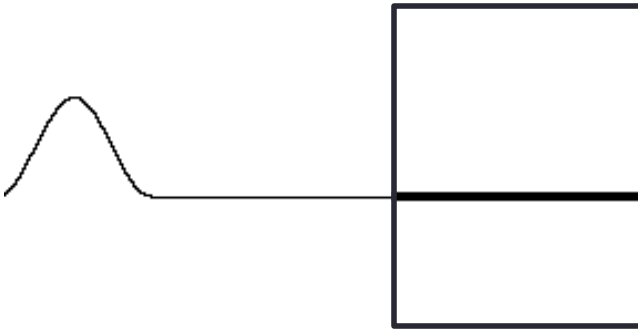


Standing wave

Various Kinds of Waves

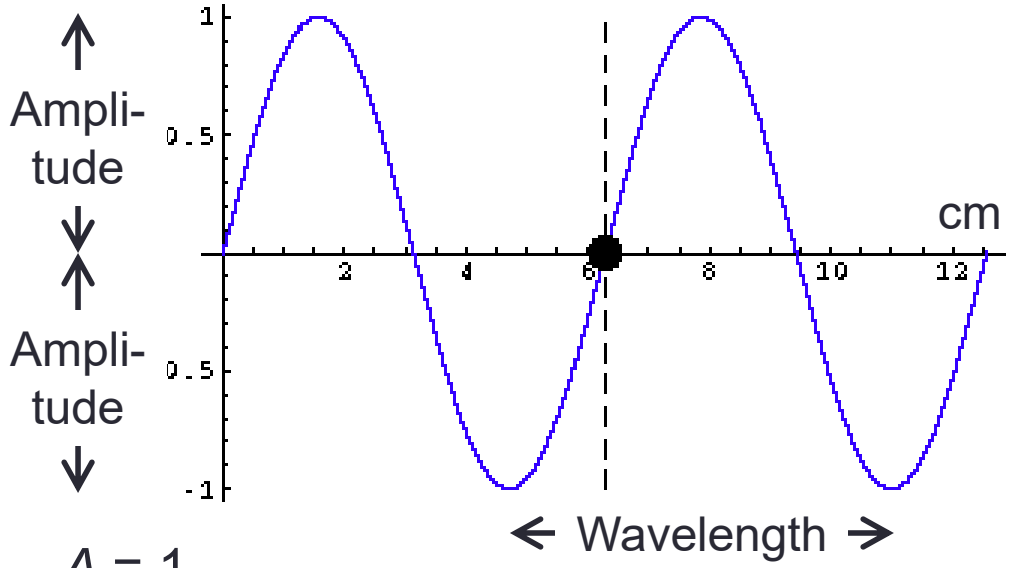


Wave packet



Reflection and Transmission

Wave: Properties



Amplitude 振幅 $A = 1$

Wavelength 波长 $\lambda = 6.3$ cm ↔ Wavenumber 波数 $\tilde{\nu} = \lambda^{-1} \approx 0.16$ cm^{-1}

Period 周期 $T = 3.0$ s ↔ Frequency 频率 $\nu = T^{-1} \approx 0.33$ Hz

Speed 波速 $s = \lambda / T = \lambda\nu \approx 2.1$ cm·s^{-1}

$c_{vacuum} = 3.0\times10^{8}$ m·s^{-1}

For visible light, $\lambda = 400$–700 nm,

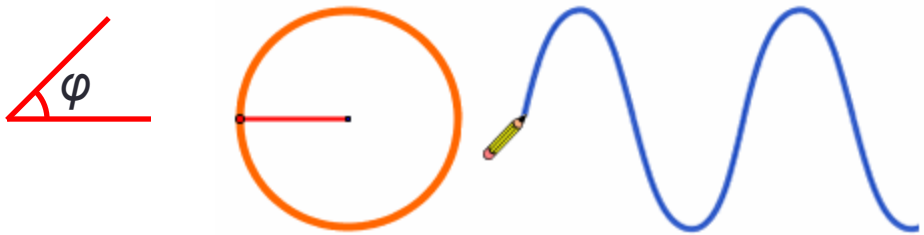
Wave: Phase 相位

Moon phases:

0 $\pi/2$ π $3\pi/2$ 2π

φ

Wave phases:



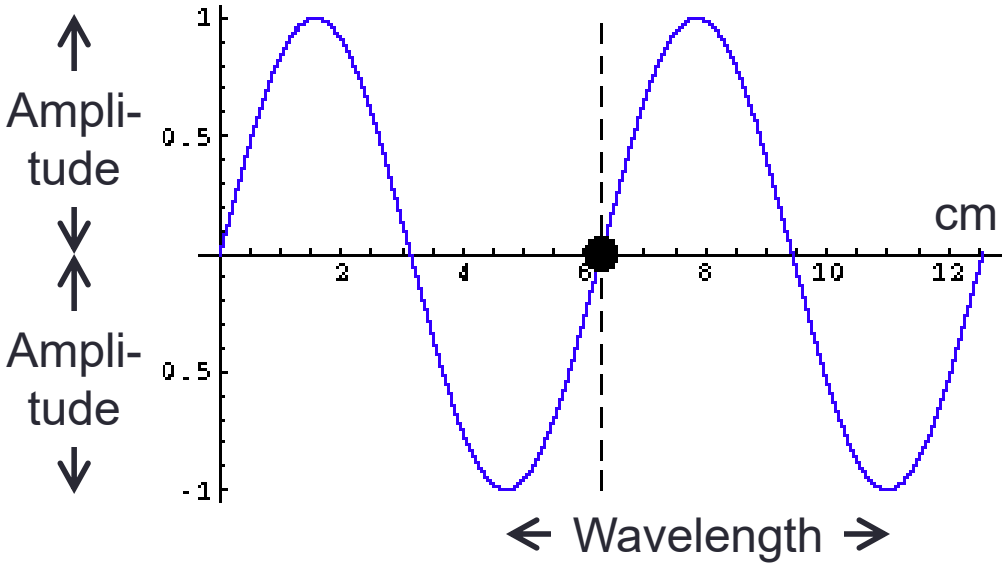
$$y = A \sin\varphi$$

Phase in Traveling Waves 行波

Phase propagates in both time and space.

$$\varphi(x) = \frac{2\pi x}{\lambda} = 2\pi\tilde{\nu}x$$

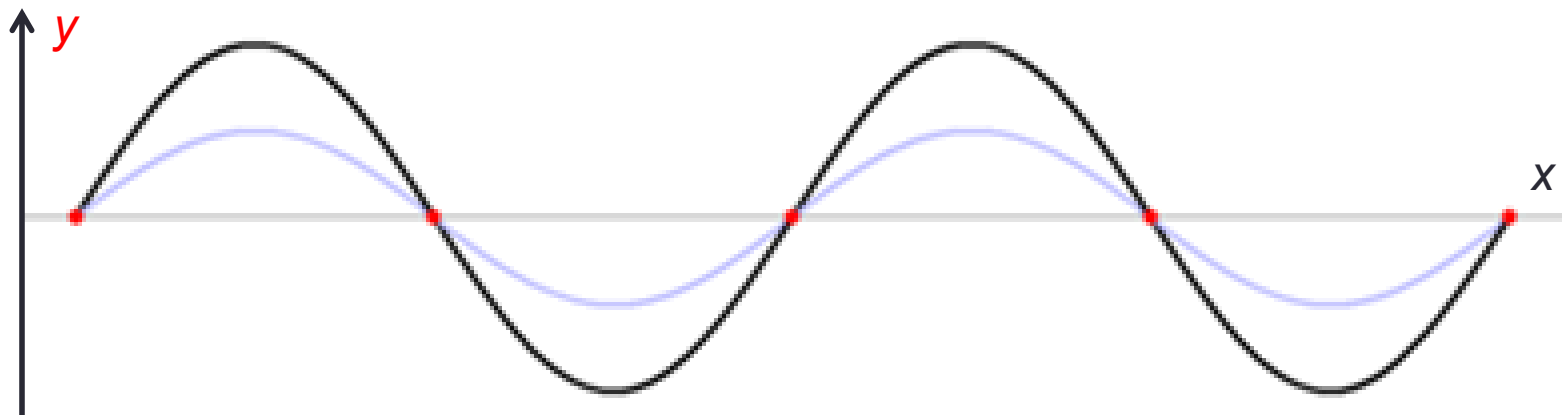
$$\varphi(t) = -\frac{2\pi t}{T} = -2\pi\nu t$$



$$y(x,t) = A \sin[\varphi(x) + \varphi(t)] = A \sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right].$$

$$\text{speed} = \frac{\Delta x}{\Delta t} = \frac{\lambda}{T}$$

Demo: Standing Waves 驻波

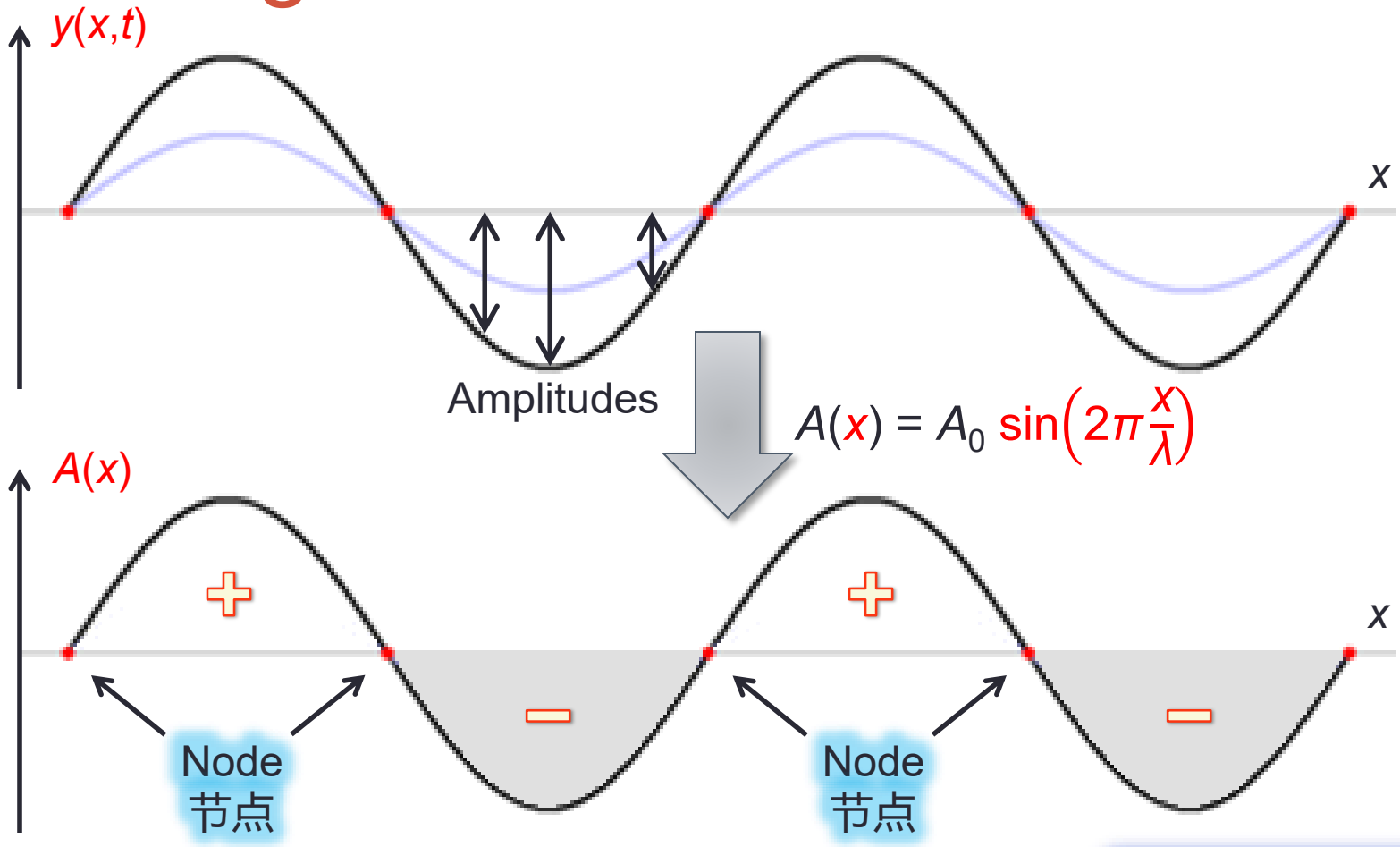


- Equals the sum of two traveling waves in opposite directions;
- Does not propagate in either direction and fits in a confined space;

$$y(x,t) = \frac{A_0}{2}\sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right] + \frac{A_0}{2}\sin\left[2\pi\left(\frac{x}{\lambda} + \frac{t}{T}\right)\right]$$

$$y(x,t) = A_0 \sin\left(2\pi\frac{x}{\lambda}\right)\cos\left(2\pi\frac{t}{T}\right).$$

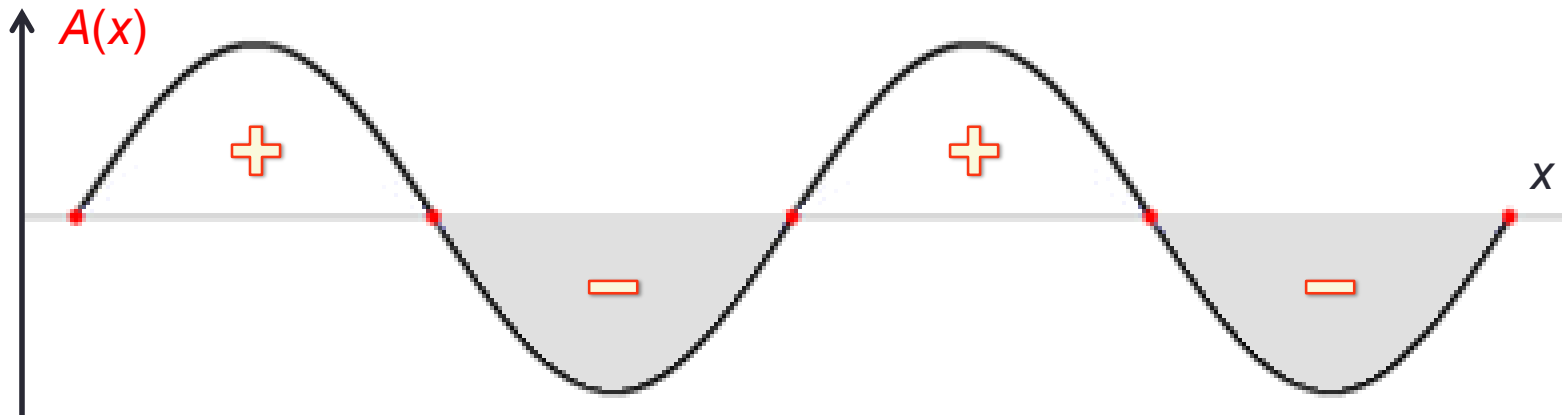
Standing Wave: Nodes



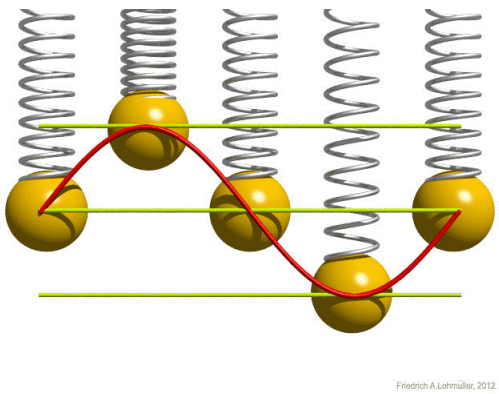
- Changes its amplitude as position varies;
- A position with a zero amplitude is called a node.

Stationary State
定态

Energy and Wave Function (1)



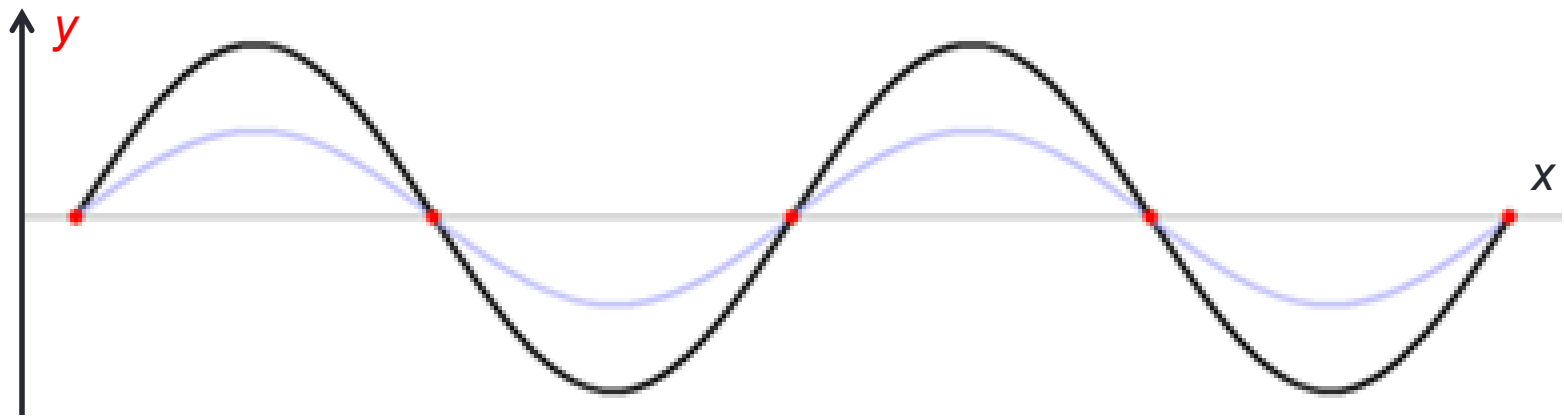
$E = \frac{1}{2}k[A(x)]^2$ is time-independent.



Let wave function 波函数 $\psi(x) = \sqrt{\frac{k}{2}}\,A(x)$, then

$$E(x) = [\psi(x)]^2$$

Energy and Wave Function (2)



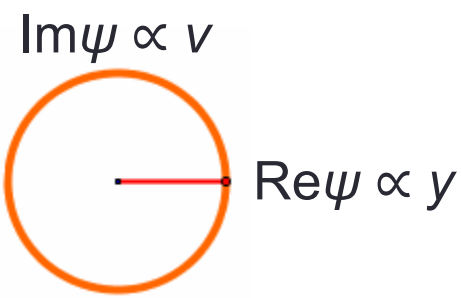
Imaginary unit
虚数单位

Generally, $E = \frac{1}{2}ky^2 + \frac{1}{2}mv^2$.

Let complex wave function $\psi(x,t) = \sqrt{\frac{k}{2}}\, y(x,t) + \mathrm{i}\sqrt{\frac{m}{2}}\, v(x,t)$, then

$$E(x) = |\psi(x,t)|^2$$

$$\psi(x,t) = \psi(x,0)\exp[\mathrm{i}\varphi(t)]$$



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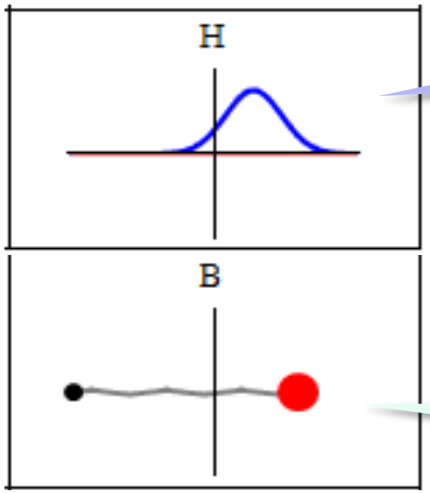
Wave Function for Electrons

- A function of position x and time t
- Has a real part and an imaginary part, in general
- Is governed by the Schrödinger Equation (1926)
  薛定谔方程

Erwin Schrödinger
(Zürich, 1887–1961)

Real
Imaginary



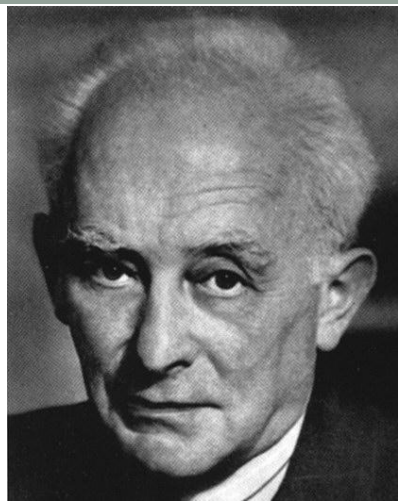
Follows
the Schrödinger Equation

Follows
Newton's Laws

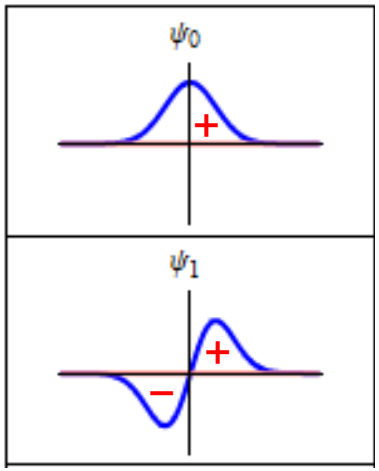
Interpretation of Wave Functions

Probability: $P(x) = |\psi(x)|^2$

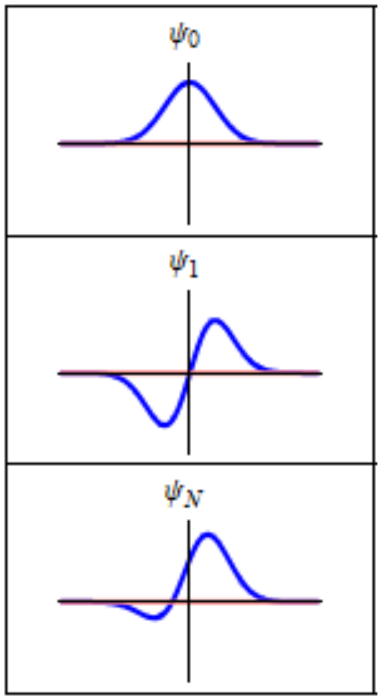
Normalization: $\int |\psi(x)|^2 \, dx = 1$
归一化



Max Born
(Göttingen,
1882–1970)



Textbook notation



Stationary
states
定态

Time-
dependent
state

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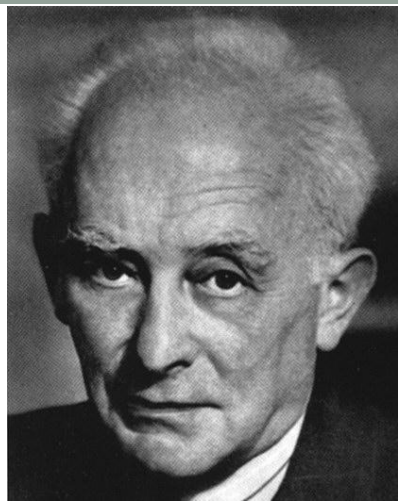
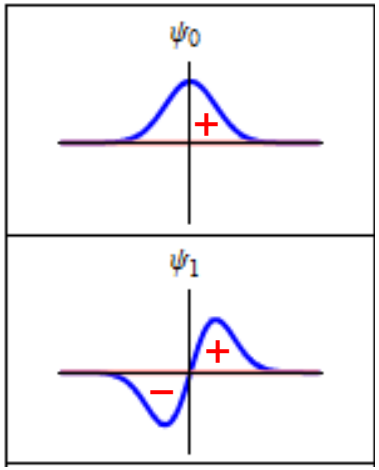
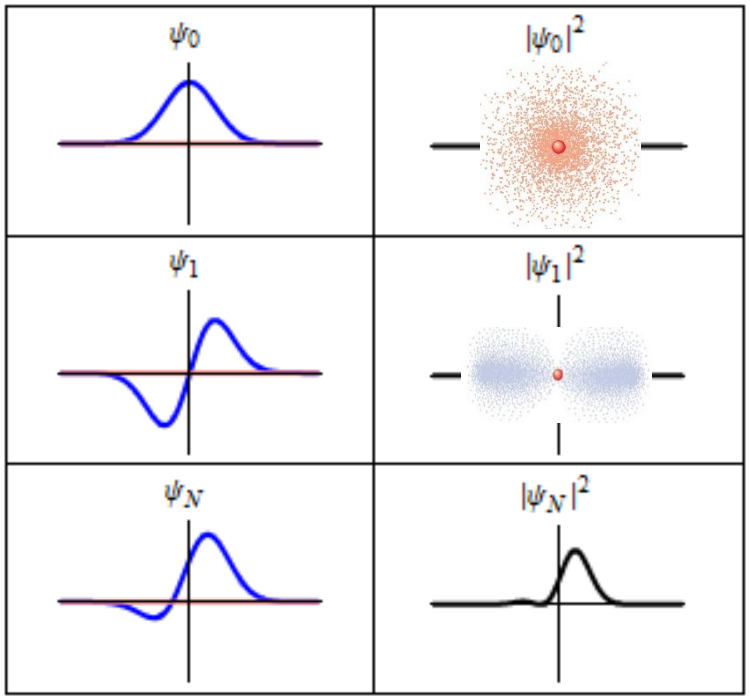
Normalization: $\int |\psi(x)|^2 \, dx = 1$
归一化



Max Born
(Göttingen,
1882–1970)



Textbook notation



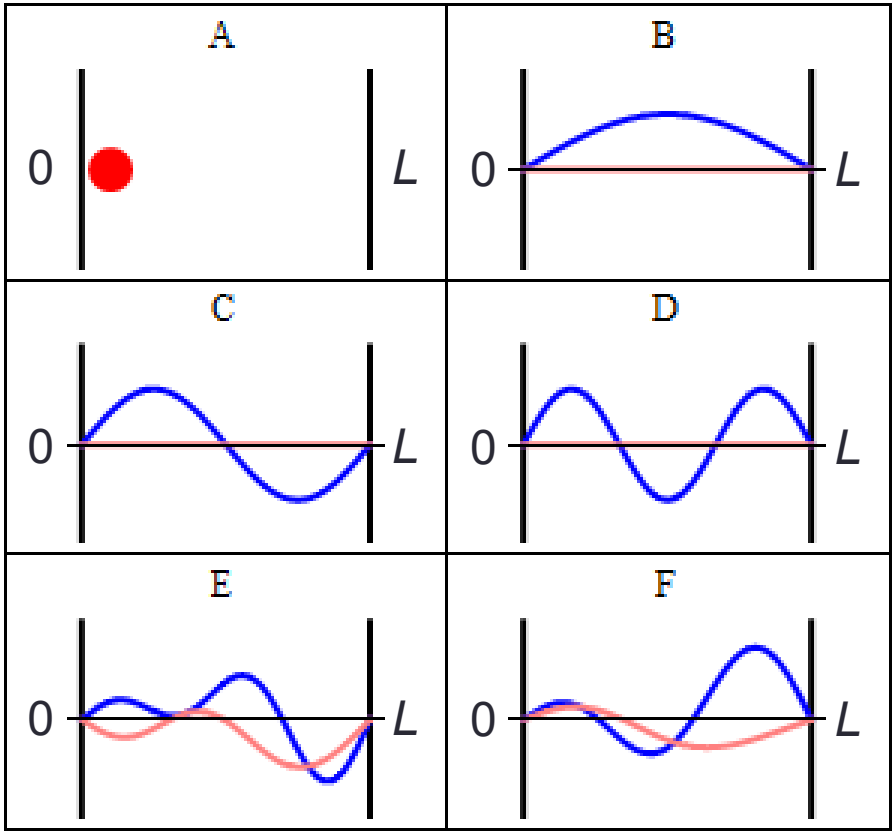
Stationary
states
定态

Time-
dependent
state

Particle in A Box

(1) $\psi(x) = 0$ for $x \leq 0$ or $x \geq L$ (Boundary condition 边界条件)

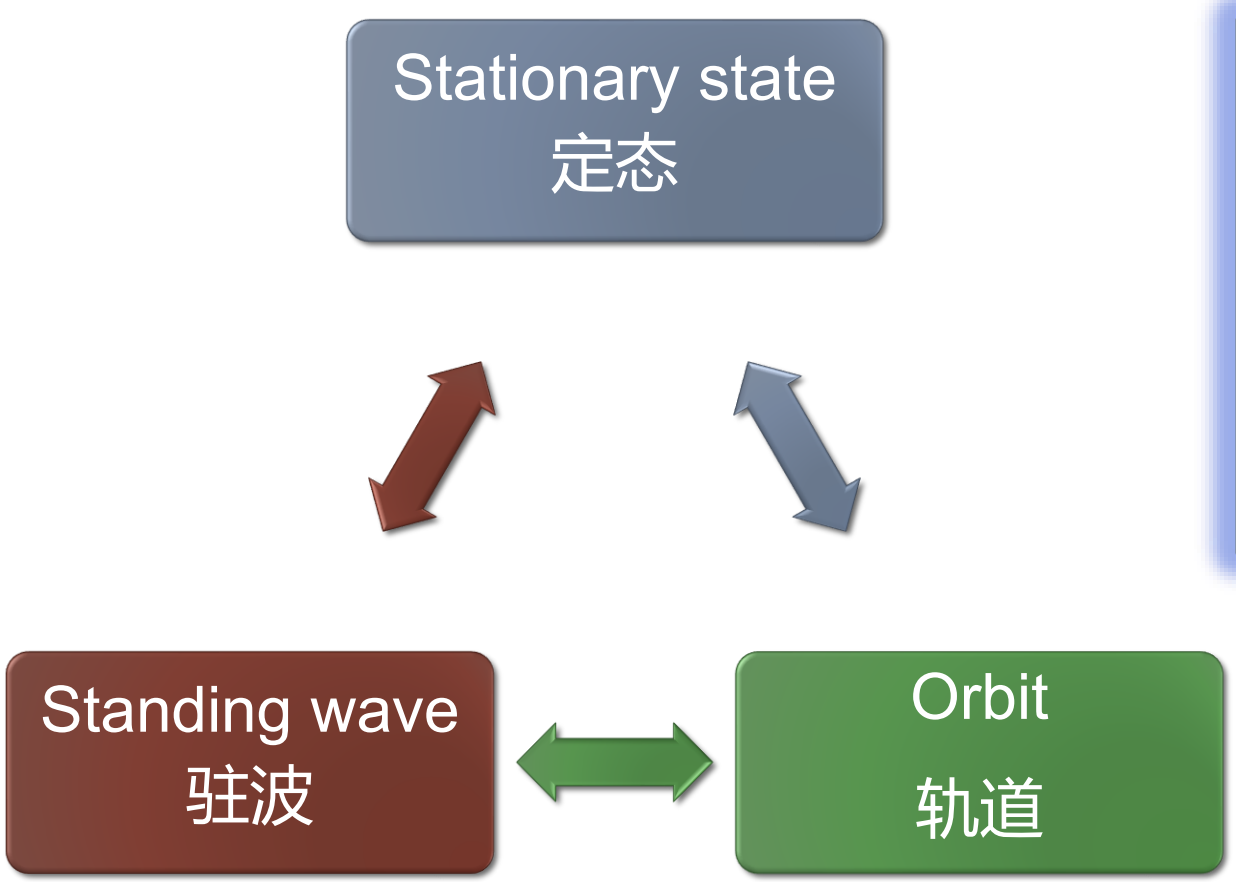
(2) $\int |\psi(x)|^2 \, dx = 1$



Stationary states 定态

Time-dependent state

Summary



Energy

Frequency

Nodes

Quantum number

Summary

$$E_{\text{photon}} = h\nu$$

- Wave–particle duality
  - Light can behave like particles
  - Electrons can behave like waves

$$p_{\text{photon}} = \frac{h}{\lambda}$$

$$\lambda_{\text{electron}} = \frac{h}{p}$$

- Wave functions
  - Describes the motion of both light and electrons
  - Its square equals probability
  - Its number of nodes indicates energy

$$P(x) = |\psi(x)|^2$$

Quantum number = Number of nodes

Next lecture series: The hydrogen atom

Reading: OGB8 §5.1, YY §2.3